# Arsalan Ali

## Mechanical Engineering / Quality / Manufacturing / Applied Software

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Texas A&M mechanical engineering student and Daikin Quality Engineering Intern connecting hands-on manufacturing, mechanical design, quality investigations, data analysis, and internal software tools.

## PROFESSIONAL EXPERIENCE

### Quality Engineering Intern

Daikin Comfort - Waller, Texas | May 2026-present

- Developed a Product Quality 0D-8D workflow and playbook foundation, clarifying investigation stages, evidence, decisions, and cross-functional ownership.
- Created a reusable Master File template for investigation background, warranty analysis, testing, supplier findings, actions, and closure documentation.
- Developing a Flask-based quality engineering beta workspace for PQR investigations, 8D content, tasks, evidence, weekly updates, and NPD checklists.
- Built and extended a GIB dashboard with shared Azure-hosted datasets, authorized publication, version history, rollback, and investigation workspaces.
- Worked with Snowflake SQL, ThoughtSpot, and dashboard prototypes on warranty analytics and reporting validation; developed quality knowledge and AI-retrieval prototypes.

### Quality Engineering Intern

Professionalized Products & Services, Inc. (PPSI) | May-December 2025

- Supported PCBA DFM, first articles, NPI, and production through Gerber, centroid / XY, BOM, and schematic review.
- Developed 5+ Mirtec MV-3 AOI programs and reorganized inspection workflow, increasing inspection throughput by 100 boards per day.
- Designed three 3D-printed SolidWorks fixture revisions for Takaya APT-9600 testing, reducing electrical false failures by 30% across three board families.
- Implemented a sensor-related corrective taping method associated with a 5% RMA reduction; documented work instructions and QA acceptance criteria.
- Developed a Python BOM comparison script / prototype and evaluated packaging material, labor, and ESD-handling tradeoffs.

### Makerspace Instructor & Technician

IDEAStudio Makerspace - Houston, Texas | September 2024-May 2025

- Trained 200+ students on 3D printing, laser cutting, woodshop tools, and CNC equipment; supported a space serving 80+ students weekly.
- Guided CAD-to-part workflows and performed calibration, troubleshooting, and preventive maintenance.

## EDUCATION

**Texas A&M University** - B.S. Mechanical Engineering; Computer Science minor | May 2028

Coursework includes thermodynamics, solid mechanics, differential equations, and engineering measurement. Earlier dual-enrollment study at Houston Community College.

# Beyond the resume.

Selected engineering projects, tools, and professional development

Explore 26 detailed case studies at arsalanali01.github.io. Employer work is summarized without internal data or proprietary source code.

## SELECTED PROJECT DEPTH

### Quality Engineering Platform

Beta application connecting investigations, 8D disciplines, teams, actions, evidence, and NPD checklists. Python / Flask, SQLAlchemy, SQLite, Alembic, Azure, and regression testing.

### GIB Investigation Dashboard

Implemented investigation views, filters, exports, dataset publication, version history, and rollback. Continuing work includes runtime performance and workspace usability.

### Warranty Intelligence & Quality AI

Warranty prototype and analytical-validation work using Snowflake / ThoughtSpot; knowledge-repository and Copilot prototypes plus isolated Azure AI Search retrieval groundwork.

### ISS Mimic Design & Mentorship

Truss-connection and solar-array mechanism redesign work, plus Spring 2025 mentor-crew instruction for Olle Middle School learners through West Houston Institute / HCC.

### 3D-Printed Rack-and-Pinion Lifter

Iterated a mechanism around a 50 g target: changed effective gearing from 2:1 to 5:1, increased axle diameter from 3 mm to 5 mm, and improved retention, alignment, and fit.

### Six Sigma DMAIC Simulation

Individual Minitab training project using a 300-sample dataset, Pareto analysis, fishbone diagrams, 5 Whys, and a control plan. Reported improvements are simulation results.

### PCB & Controls Studies

Pulse-oximeter and open-source tracker PCB design studies; simulated conveyor ladder logic and structured-text batch sequencing. These are design / simulation projects.

## TOOLS & METHODS

**Quality & manufacturing:** 8D, RCA, CAPA, FMEA fundamentals, DFM / DFA, FAI / NPI, AOI, flying-probe testing, SMT / PCBA, BOM review, work instructions, acceptance criteria, ESD control.

**Software, data & cloud:** Python, Flask, JavaScript, HTML / CSS, SQL, Snowflake, ThoughtSpot, Excel, Minitab, Git / GitHub, GitHub Actions, pytest, SQLAlchemy, SQLite, Azure App Service / Storage / AI Search.

**Design, measurement & workflow:** SolidWorks, Fusion 360, AutoCAD, KiCad, Altium Viewer, GerbView, GD&T fundamentals, 3D printing / Cura, LabVIEW / DAQ, SharePoint, Lists, Power Automate, Office Scripts, Microsoft Copilot.

## LEADERSHIP & CREDENTIALS

**Autodesk Design & Make Ambassador** - Texas A&M, June 2025-present. Peer CAD support and student design-and-make outreach. **NASA NCAS** - aerospace educational program, January-June 2025. **Chevron Academy Mentoring Program** - professional development.

**Selected credentials:** AIGPE Lean Six Sigma Yellow Belt (May 2025); Git & GitHub Bootcamp (August 2025); LinkedIn Data Analytics career-skills course (July 2025); New Space Community of Experience (June 2025); Python programming and integrated ISO management training. Credential links are on the portfolio.